Calculus I

Computation problems #5

Topics:

- Extreme values of functions
- The Mean Value Theorem
- The first derivative test
- Concavity and curve sketching

In Problems 1 and 2, find the absolute maximum and minimum values of each function on the given interval. Then graph the function. Identify the points on the graph where the absolute extrema occur, and include their coordinates.

Problem 1. $f(x) = x^2 - 1, \quad -1 \leq x \leq 2.$

Problem 2. $g(x) = \sqrt{4 - x^2}, \quad -2 \leq x \leq 1.$

In Problems 3 and 4, find the extreme values of the function and indicate where they occur.

Problem 3. $y = \sqrt{x^2 - 1}$.

Problem 4. $y = \dfrac{1}{\sqrt[3]{1 - x^2}}$.

In Problems 5, 6 and 7:

(a) Find the intervals on which the function is increasing and decreasing.

(b) Then identify the function's local extreme values, if any, saying where they are taken on.

(c) Which, if any, of the extreme values are absolute?

(d) Support your findings with a graphing calculator or computer grapher.

Problem 5. $g(t) = -t^2 - 3t + 3$.

Problem 6. $f(\theta) = 3\theta^2 - 4\theta^3$.

Problem 7. $h(x) = \sqrt[3]{x}(x^2 - 4)$.

Identify the inflection points and local maxima and minima of the given functions in Problems 8-10. Identify the intervals on which the functions are concave up and concave down.

Problem 8. $y = \dfrac{x^3}{3} - \dfrac{x^2}{2} - 2x + \dfrac{1}{3}$.

Problem 9. $y = \dfrac{3}{4}(x^2 - 1)^{2/3}$.

Problem 10. $y = x + \sin 2x, \quad -\dfrac{2\pi}{3} \leq x \leq \dfrac{2\pi}{3}$.

. . . . . .

Answer to selected problems

4. No absolute maximum and minimum.

10. Inflection points $x = \pm\dfrac{\pi}{2}$ and $x = 0$; Local maximum $y(-\dfrac{\pi}{3}) = -\dfrac{\pi}{3} - \dfrac{\sqrt{3}}{2}$, local minimum $y(\dfrac{\pi}{3}) = \dfrac{\pi}{3} + \dfrac{\sqrt{3}}{2}$.